

4TH YEAR ADVANCE LINUX -PRACTICALS

Modified 04/14/2026 13:44:04 Created 03/26/2026 08:24:28

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Module 1: Foundation & Environment Setup

Theory

- Server distribution selection: CentOS Stream, RHEL, Rocky Linux, AlmaLinux, Ubuntu etc
- Installation methods: minimal install, network install, Kickstart basics
- Essential pre-configuration: hostname, static IP, SELinux states, firewall overview

Practical Labs

- Install Linux Server Of your choice preferably Ubuntu, Cent OS
- Install a client machine of your choice
- Configure IP addresses
- Initial hardening: disable unnecessary services/have a firewall
- SSH server installation

Objective:

Complete OS installation of a server and a client PC(Use Virtualization preferably virtualbox, VMware ..). Client and the Server must ping each other

-Server Remote access through ssh

Module 2: DNS Server Administration (BIND)

Theory

- DNS fundamentals: hierarchy, zones, record types (A, AAAA, CNAME, MX, NS, SOA, PTR, TXT, SRV)
- BIND architecture: named.conf structure, zones, views, ACLs
- DNS security: DNSSEC concepts, TSIG for zone transfers, response policy zones (RPZ)

Practical Labs

- Install BIND 9(`bind`, `bind-utils`)
- Configure Master DNS server with forward and reverse zones
- Create multiple zone files for different domains
- Configure Slave DNS server with zone transfers (AXFR/IXFR), TSIG authentication
- Testing: `dig`, `nslookup`, `rndc`, zone file validation with `named-checkzone`

Objective: Fully Qualified domain working when tested with dig or nslookup

Module 3: SSH, FTP & Secure File Transfer

Theory

- FTP protocol: active vs. passive mode, command/data channels, security limitations
- SFTP vs. FTPS: SSH File Transfer vs. FTP over SSL/TLS
- vsftpd, ProFTPD configuration models

Practical Labs

- Install and configure vsftpd or ProFTPD OR any other linux FTP Server

Objectives

Deploy FTP server with chroot jails; configure SFTP as secure alternative, anonymous access (disabled), local user access, chroot jails.

Module 4: Web Server Administration

Theory

- HTTP/HTTPS fundamentals: request/response cycle, status codes, virtual hosting concepts
- SSL/TLS: certificate types, Let's Encrypt, cipher suites, HSTS

Practical Labs

- Install and Configure Apache, httpd or Nginx

Objective:

Accessible website from the Client PC

Module 5: Email

Theory

Email architecture deep dive: MUA (Mail User Agent), MSA (Mail Submission Agent), MTA (Mail Transfer Agent), MDA (Mail Delivery Agent)

- SMTP protocol: envelope vs. headers, relaying, authentication, STARTTLS
- Procmail: mail filtering, delivery rules

Practical Labs

Configure Postfix to accept local SMTP mail.

Configure Dovecot to allow a user to login via IMAP.

Install Roundcube (requires Apache/Nginx and PHP) and point it to localhost for IMAP/SMTP.

Objective:

Two emails communicating



Practical Assessment Timeline: 4th Year Advanced Linux

Week	Date (Thursday)	Module	Topic	Key Deliverables
Week 1	April 23, 2026	Module 1	Foundation & Setup	Functional Server & Client (Pingable); SSH Access.
Week 2	April 30, 2026	Module 2	DNS Admin (BIND)	Master/Slave DNS; Successful Forward/Reverse Lookups.
Week 3	May 7, 2026	Module 3	SSH & FTP Services	FTP,FTPS and SFTP
Week 4	May 14, 2026	Module 4	Web Administration	Virtual Hosting (Apache/Nginx); Accessible via DNS Name.
Week 5	May 28, 2026	Module 5	Email Systems	Postfix/Dovecot/Roundcube; Successful Email Exchange.

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